



Best-Self Digital Twin

Counterfactual simulator for clinically plausible future selves from longitudinal biomarkers.

FUTURES

12,000

PARETO SET

48

SELECTED PATH

Maximum

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Member Twin

State is built from repeated lab panels.

Synthetic member

M-1042 - Metabolic ApoB riser

10

Metabolic ApoB riser

46, Female | M-1042 | synthetic

ApoB rising

insulin high

ALT drift

APOB

112

+26 mg/dL drift

FASTING INSULIN

16.1

+6.6 uIU/mL drift

HS-CRP

2.4

+1.1 mg/L drift

ALT

34

+14 U/L drift

OMEGA-3 INDEX

4.6

-0.6 % drift

Objective

Pick the future self to optimize for.

Best overall

Lowest burden

No-med path

Fast ApoB drop

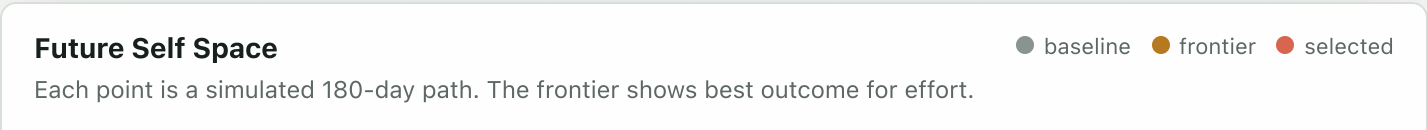
Metabolic reset

Maximum change

Medication pathways allowed

Keep burden low

Generate 12,000 futures

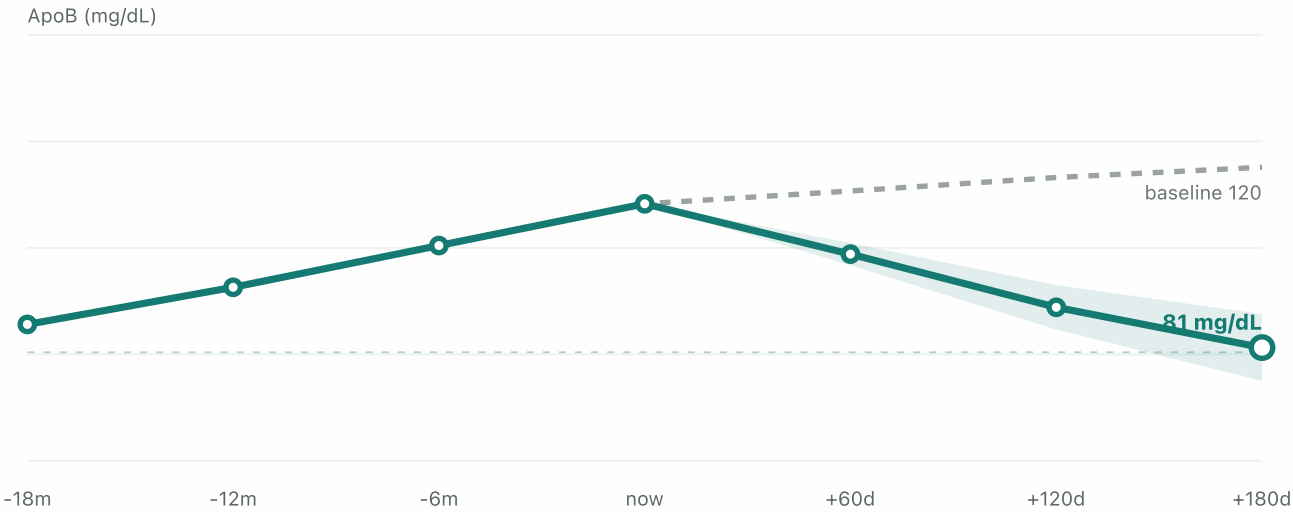




Biomarker Forecast

Selected path versus no-change baseline with uncertainty.

ApoB



ApoB: -31

Fasting insulin: -6.9

hs-CRP: -1.3

ALT: -10

Omega-3 index: +1.6

Tradeoff Frontier

The simulator does not choose one "right" future. It exposes options.

TOP FRONTIER SCORE

45

Maximum, effort 89

LOW-BURDEN OPTION

18

Metabolic, effort 44

SELECTED CONFIDENCE

63%

From cohort density and intervention complexity

TIME TO SIGNAL

42d

First likely measurable lab movement

Pareto framing: the selected path buys 45 points of biomarker improvement for 89 effort. A user can trade speed, medication preference, and adherence burden without pretending one path is universally best.

Choose A Future Self

Top paths under the current objective.

1. Best match for objective

Metabolic + Strength training | effort 89

45

2. Low-friction self

Omega-3 + nutrition + Sleep + alcohol reset | effort 46

19

3. No-med self

Metabolic + Sleep + alcohol reset | effort 61

29

4. Fast ApoB self

Lipid + Metabolic | effort 86

42

5. Metabolic reset self

Metabolic + Sleep + alcohol reset | effort 84

44

Selected Simulation

Expected changes after 180 days.

BEST-SELF SCORE

45

BURDEN

89

ApoB

-31

81 mg/dL forecast

Fasting insulin

-6.9

9.2 uIU/mL forecast

hs-CRP

-1.3

1.1 mg/L forecast

ALT

-10

24 U/L forecast

Omega-3 index

+1.6

6.2 % forecast

Twin Reasoning

Latent state: ApoB rising, insulin high, ALT drift. The twin maps recent slopes, volatility, and cross-marker coupling into a trajectory state.

Selected future: Maximum path using Metabolic + Strength training. Forecast gain is 45 with 63% confidence.

Closed loop: retest in 90-180 days; prediction error recalibrates the member twin and improves cohort-level counterfactuals.